

National College of Ireland

Higher Diploma in Science in Computing, Year 1, HDCSDEVINTJAN25I

Higher Diploma in Science in Computing, Year 1, HDSDEV_SPOL24_Y1

Higher Diploma in Science in Computing, Year 1, HDSDEV_JAN25

Release Date: Wednesday, 12 February, at 09:00

Submission Deadline: Part-1: Saturday, 15 March at 23:55

Part- 2: Saturday, 26 April at 23:55

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Object-Oriented Software Engineering

Continuous Assessment (CA) Type: Project 100%

Instructions:

- Marks for Object-Oriented Software Engineering will be assessed by a single project, split into two parts (Part-1 and Part-2). Each part is worth 50% of the total mark.
- Students may work **individually or in teams of up to two**.
- The task requires designing an application for a particular business domain determined by the **last digit of your student number**, as outlined in the next table. Selecting the business domain for a team depends on the team member whose surname starts with the letter closest to the **beginning of the alphabet**. Alternatively, a team member may propose a familiar business domain, subject to discussion with the lecturer and approval before commencing work on the assessment.

Last Digit of Student Number	Business Domain
0 or 1	An art dealer or a beauty salon
2 or 3	An online take away restaurant or a gym
4 or 5	A dental clinic management system or a pharmacy
6 or 7	An estate agent or a driving license renewal
8 or 9	A cinema ticket booking system, a motor mechanic system

For Example, if your student ID number is x19876¹, the business domain you may choose is either An art dealer or a beauty salon.

SUBMISSION DETAILS:

- The report should be uploaded as two separate documents in **PDF format** for both parts (Part-1 and Part-2) to the **Turnitin link**.
- For Part-2 only, you'll also need to upload your code artifact. This should be a zip file containing the developed classes and the suite of tests conducted.
- In each part, there are tasks that can be completed together as a team or separately by team members. **All answers for individual tasks should be in one report, with each team member's contribution clearly marked.**

For example:

Use Case Description

Student Number for Team Member 1

Answer from team member 1..

Student Number for Team Member 2

Answer from team member 2...

- The submission for both parts should be made by only **one nominated team member**. All the members' names should be mentioned on the first page of the report.
- **Penalty will not be applied for the Late Submissions**, if the student applied for an extension through [NCI360](#) and it was approved.
- Please note that the lecturer cannot grant extensions. Extensions can only be granted if approved after the submission of a personal circumstances form. Please consult NCI360 or programme coordinator for further information.
- Any written work created by others must be properly cited and should be paraphrased or summarized where possible, otherwise it should be included in quotes. Figures not created by you should include an acknowledgment detailing the name(s) of the creator(s). Small code snippets found on the internet should not be claimed as your own, but instead a comment should be included in the source code indicating where you obtained it.
- All **submissions will undergo electronic screening using Turnitin database** to detect any signs of academic misconduct, such as plagiarism, collusion, or misrepresentation.
- All work submitted to NCI is subject to [NCI's Guidelines on Academic Integrity](#). Every student must visit the Academic Integrity webpage. **Not knowing its content will not be accepted as an excuse if suspected of academic misconduct.** Any student suspected of academic misconduct will be referred to the Disciplinary Committee. **Including the AI declaration Project Cover Sheet is mandatory in your submission document and can be downloaded, [here](#).** Please explicitly declare any AI involvement in your submission by using the Project Cover Sheet and furnishing the AI's prompts.
- Where AI generated content is presented, **the prompts given to the AI must be included in an appendix. The main body of the assignment must be your original work.** Failure to comply with these guidelines violates our academic integrity policy. Remember to utilize technology responsibly, as a tool to enhance, not replace, your creative and critical thinking.

Module Learning Outcomes

On successful completion of this module the learner will be able to:

LO1	Demonstrate the conceptual, practical and technical skills of planning and monitoring a project plan using an appropriate CASE tool.
LO2	Describe in detail the theory, concepts and methods pertaining to Software Engineering such as Agile and UML.
LO3	Create requirements using use-case modelling concepts.
LO4	Demonstrate conceptual and technical skills in the analysis, design, implementation and test of a software development solution individually or as part of a team.
LO5	Employ tools and techniques for Software Engineering,

All learning outcomes (LOs) are assessed in the first part of the assessment. LOs 2, 4 and 5 are assessed in the second part of the assessment.

Part 1 (50% weight)

In this first part of the assessment, you're required to submit a report that contains the tasks outlined below. These tasks include introducing and planning a project, and applying UML diagrams to the selected business domain specified in the table provided earlier. **UML diagrams should be created exclusively using UML software tools.** Detailed descriptions of the tasks and marking schemes are provided below.

Task	Task description	Marks	Maximum word limit
1.	As a team, identify and describe the actors, and construct a use case diagram. Briefly explain what your use case diagram illustrates in your own words.	15	600
2.	For each team member, should select a distinct use case from the use case model and describe it in your own words. Ensure that each description includes an alternate or exceptional flow. Explain why you chose this use case.	15	600
3.	As a team, create a Conceptual Class Diagram for the chosen use case in Task 2. The diagram should effectively represent the system's structure and include at least three or more of the following key elements: (1) attributes, (2) relationships, (3) navigability, (4) association, (5) multiplicity, and (6) composition. Describe the information conveyed by the conceptual class diagram and its significance in understanding the system's structure and relationships.	20	No limit
4.	Create a glossary that lists and defines all the project related terms that require clarification.	5	200
5.	As a team, create a System Sequence Diagram (SSD) for one of the distinct use cases described in Task 2. The diagram should illustrate the flow of interactions between the actor (user) and the system, showing how the system responds to external inputs.	15	500
6.	Each team member should develop a contract for one of the system operations in the sequence diagram.	15	No limit

7.	Create individual communication diagrams based on the specified contracts, ensuring that each diagram prominently showcases the implementation of named design patterns. (Note that each team member should come up with their own communication diagram depending on the contracts chosen in the previous task 6.)	15	No limit
100			

Using the following scale, each task in this section will be awarded a percentage of the total mark.

Mark Range	Criteria
Fail < 40%	A poor answer that demonstrates little or no understanding of the theory and practice of Object Oriented Software Engineering and the use of UML.
Pass 40-49%	An adequate answer that demonstrates a somewhat limited understanding of the theory and practice of Object Oriented Software Engineering and the use of UML.
H2.2 50-59%	A good answer that demonstrates a reasonable understanding of theory and practice of Object Oriented Software Engineering and the use of UML.
H2.1 60-69%	A very good answer that demonstrates a complete understanding of the theory and practice of Object Oriented Software Engineering and the use of UML.
H1 ≥ 70%	An excellent answer that demonstrates an exceptionally in depth understanding of the theory and practice of Object Oriented Software Engineering and the use of UML.

Part 2 (50% weight)

For the second part of the assessment, you need to submit a report covering the tasks outlined below. These tasks involve the application of agile methodologies like Scrum, Kanban and TDD to the selected business domain, along with the development and testing of a crucial component of your project. Detailed descriptions of the tasks and the associated marking schemes are provided below.

Task	Task description	Marks	Maximum Word limit
8.	Propose and justify the agile development methodology you will follow in your own words, highlight its differences from the Waterfall approach, particularly in terms of project structure, adaptability, and risk management.	25	700
9.	Submit comprehensive artifacts documenting the Agile methodology used, including user stories, backlogs, and burndown charts, providing detailed insights into the project's progress and objectives.	25	700
10.	Identify potential risks specific to your project and describe the quality and communication management strategy.	15	No limit
11.	Develop all the classes required to implement one of the use cases described in part 1 of the assessment in a suitable OO language (e.g., Java or C++).	10	No limit
12.	Create test cases or scenarios to test the classes developed above, naming and justifying the test methodology followed. Describe the tests carried out, detailing how you will ensure that the classes are free from errors and detail the results of the tests.	25	No limit
		100	

Using the following scale, each task in this section will be awarded a percentage of the total mark.

Mark Range	Criteria
Fail < 40%	A poor answer that demonstrates little or no understanding of the implementation of project management approaches, agile methodologies and software testing methods.
Pass 40-49%	An adequate answer that demonstrates a somewhat limited understanding of the implementation of project management approaches, agile methodologies and software testing methods.

H2.2 50-59%	A good answer that demonstrates a reasonable understanding of the implementation of project management approaches, agile methodologies and software testing methods.
H2.1 60-69%	A very good answer that demonstrates a complete understanding of the implementation of project management approaches, agile methodologies and software testing methods.
H1 ≥ 70%	An excellent answer that demonstrates an exceptionally in depth understanding of the implementation of project management approaches, agile methodologies and software testing methods.

Software Tools

- Visual Paradigm software tool is recommended. NCI has a license for the professional version of Visual Paradigm, please ask your lecturer for more information. NetBeans is the preferred tool for the unit testing of classes. Trello or Jira free tools or any other tool of your choice can be used for Project Management.

Please Note: Your lecturer reserves the right to request a one-to-one viva presentation with any student should this be deemed necessary for any reason.